

Fair, Interpretable, and Actionable Readmission Risk Modeling — Diabetes 130-US Hospitals: Data Preparation and Exploratory Data Analysis (Part 1)

1. Title and Brief Introduction

Title: Fair, Interpretable, and Actionable Readmission Risk Modeling — Diabetes 130-US Hospitals: Data Preparation and Exploratory Data Analysis (Part 1)

Introduction: This report presents the first phase of a comprehensive project aimed at understanding and predicting hospital readmission among diabetic patients. Part 1 focuses on data quality assessment, data cleaning, preprocessing, exploratory data analysis, and statistical investigation of factors associated with readmission.

The original dataset contains 101,766 hospital encounters and 50 variables, including demographic characteristics, admission and discharge information, medical diagnoses, medication-related variables, and previous healthcare utilization.

The primary objective is to transform the raw data into a reliable analytical dataset, identify important patterns associated with readmission, and prepare the data for the machine learning and model evaluation stages described in Part 2.

2. Objective and Relationship to Part 2

Objective of Part 1: The main objective is to assess data quality, manage missing and invalid values, construct meaningful features, examine the target distribution, and identify variables associated with readmission.

The analysis addresses three main questions:

1. What are the main characteristics of the readmission population?
2. Which variables differ most between readmitted and non-readmitted encounters?
3. Which variables show the strongest statistical associations with readmission?

Relationship to Part 2: Part 1 establishes the data preparation and exploratory analysis foundation. The cleaned and encoded datasets produced in this phase are intended to serve as inputs for predictive modeling, model comparison, threshold selection, interpretability, error analysis, and subgroup fairness evaluation in Part 2.

3. Methodology and Workflow

The analytical workflow was implemented in Python using data manipulation, statistical analysis, and visualization libraries. The main stages were as follows.

3.1. Initial Data Inspection and Missing-Value Analysis

The original dataset contained 101,766 encounters and 50 variables. Missing values were examined across all variables, using an 80% missingness threshold as an initial criterion for identifying variables with excessive missing information.

The variables weight, payer_code, max_glu_serum, and A1Cresult were removed. Although payer_code did not exceed the missingness threshold, it was excluded because of its limited relevance to the analytical objective.

The remaining variables were retained or handled according to their missingness patterns and analytical relevance.

3.2. Identifier Management and Duplicate Analysis

The dataset contains encounter_id and patient_nbr. The encounter identifier was removed because it does not provide clinically meaningful predictive information. The patient identifier was saved separately in patient_ids_for_split.csv before being excluded from the analytical feature set.

The dataset contains 71,518 unique patients among 101,766 encounters. A total of 16,773 patients have multiple encounters, demonstrating the importance of accounting for patient identity during data splitting. Otherwise, encounters belonging to the same patient could appear in both training and test sets, potentially causing information leakage.

No completely duplicated records were identified, so no observations were removed because of exact duplication.

3.3. Target Definition and Record Filtering

The original readmitted variable contains three categories: NO, >30, and <30. Their respective frequencies were 54,864 (53.91%), 35,545 (34.93%), and 11,357 (11.16%).

A binary variable, `readmitted_binary`, was constructed to distinguish between non-readmitted and readmitted encounters. In the current implementation, the positive class includes both readmissions within 30 days and readmissions after 30 days.

Records corresponding to death or hospice discharge were removed, reducing the dataset to 99,343 encounters. Records with missing race values were subsequently removed, leaving 97,109 encounters. One record with an invalid gender value was also removed, resulting in 97,108 encounters.

3.4. Categorical Feature Engineering

Sparse categories were grouped to reduce unnecessary dimensionality and improve the stability of subsequent analysis. Admission types 4, 7, and 8 were combined into Other. Admission sources 8, 10, 11, 13, 14, 22, and 25 were also grouped into Other. Discharge disposition code 18 was relabeled as Unknown.

The original age categories were consolidated into three groups: under 30 years, 30–60 years, and 60 years and older. This representation provides a more compact summary of age.

The diagnosis variables `diag_1`, `diag_2`, and `diag_3` were mapped into broader clinical categories, including circulatory, respiratory, digestive, genitourinary, diabetes, neoplasms, musculoskeletal, injury and poisoning, other, and missing. The original diagnosis-code variables were replaced by their grouped representations.

3.5. Medication Variables and Numerical Data

Medication variables were examined for extremely low variability. Variables in which a single category accounted for at least 99.5% of observations were considered nearly constant and removed.

The variables `change` and `diabetesMed` were also examined to compare readmission rates across medication-related categories.

Numerical variables were evaluated using descriptive statistics, distributions, and boxplots. Several count variables, including previous outpatient visits, emergency visits, and inpatient admissions, exhibited right-skewed distributions and a high proportion of zero values.

Potential outliers were examined using boxplots and the interquartile range criterion. These observations were retained because unusually high healthcare utilization may represent genuine clinical history rather than measurement errors.

3.6. Statistical Analysis and Visualization

Numerical variables were compared between readmitted and non-readmitted encounters using descriptive statistics and the Mann–Whitney U test.

The point-biserial correlation coefficient was used to quantify the association between numerical variables and the binary target. Cramer's V was used for categorical variables.

A correlation matrix was also calculated to examine relationships among numerical predictors. Pairwise scatterplots were used to investigate visual patterns and assess whether the target groups could be clearly separated by individual numerical features.

Finally, numerical and categorical variables were ranked according to their association strength with the target.

4. Results and Analysis

4.1. Final Dataset and Target Distribution

Following data cleaning and preprocessing, the final analytical dataset contained 97,108 encounters and 32 variables, comprising 8 numerical and 24 categorical variables.

The target distribution consisted of 85,982 non-readmitted encounters and 11,126 readmitted encounters. Therefore, approximately 11.46% of the encounters belonged to the positive class, indicating substantial class imbalance.

The cleaned dataset was saved as `diabetic_cleaned.csv`. An encoded version was also created for subsequent machine learning analysis, containing 97,108 observations and 181 variables.

Figure 1. Percentage of missing values across variables.

Figure 2. Distribution of the binary readmission target.

4.2. Demographic and Clinical Characteristics

The demographic analysis showed that most encounters involved patients aged 60 years or older. Race, gender, admission type, admission source, discharge disposition, and primary diagnosis were also examined to characterize the population.

The results provide an overview of the demographic and clinical composition of the dataset and help contextualize the subsequent comparisons between readmitted and non-readmitted encounters.

Figure 3. Demographic distributions by race, gender, and age group.

Figure 4. Distributions of selected admission, discharge, and diagnosis characteristics.

4.3. Differences Between Readmitted and Non-Readmitted Encounters

The numerical comparisons indicated that previous inpatient utilization was an important distinguishing characteristic. The number of previous inpatient encounters showed the strongest numerical association with readmission.

Other variables showing differences between the groups included previous emergency visits, number of diagnoses, length of hospital stay, number of medications, and number of laboratory procedures.

Categorical variables were also examined by comparing readmission rates across their categories. These comparisons highlighted differences across age groups, race, and primary diagnosis groups.

The results suggest that readmission is associated with multiple aspects of healthcare utilization and clinical characteristics rather than a single demographic or medication-related variable.

Figure 5. Comparison of numerical variables between readmitted and non-readmitted encounters.

Figure 6. Readmission rates by age group.

Figure 7. Readmission rates by race.

Figure 8. Readmission rates by primary diagnosis group.

4.4. Association Strength and Variable Ranking

The point-biserial correlation analysis identified `number_inpatient` as the strongest numerical correlate of readmission, with a coefficient of 0.1687. Other numerical associations included `number_emergency` (0.0610), `number_diagnoses` (0.0536), `time_in_hospital` (0.0471), and `num_medications` (0.0423).

Among categorical variables, `discharge_disposition_id` showed the strongest association, with a Cramer's V of 0.1143, followed by `medical_specialty` (0.0772).

The overall ranking placed `number_inpatient` first, followed by `discharge_disposition_id`, `medical_specialty`, `number_emergency`, and `number_diagnoses`.

These findings indicate that previous hospital utilization and discharge-related characteristics contain useful information about readmission. However, association measures do not establish causation and do not necessarily represent the final importance of variables in a multivariable predictive model.

Figure 9. Correlation matrix of numerical variables.

Figure 10. Pairwise relationships among numerical variables.

Figure 11. Top 10 variables ranked by association with readmission.

4.5. Interpretation of the Exploratory Findings

The exploratory analysis suggests that readmission risk is associated with a combination of previous healthcare utilization, discharge circumstances, and clinical characteristics.

The strongest individual numerical association was observed for previous inpatient admissions, while discharge disposition had the strongest association among categorical variables. These findings provide a useful starting point for model development.

However, the observed associations are relatively limited in magnitude, and the scatterplots do not indicate a clear separation between the target groups based on individual numerical features. This suggests that prediction may require combining multiple variables and capturing nonlinear relationships.

The substantial class imbalance also means that subsequent model evaluation should consider metrics beyond accuracy, including recall, precision, F1-score, ROC-AUC, and PR-AUC.

5. Summary and Conclusions

Summary: Part 1 established a structured data preparation and exploratory analysis workflow for the Diabetes 130-US Hospitals dataset. The process included missing-value assessment, identifier management, duplicate analysis, target construction, record filtering, categorical feature engineering, medication-variable screening, numerical analysis, outlier assessment, and statistical association analysis.

The final cleaned dataset contains 97,108 encounters and 32 variables. The target remains imbalanced, with 11,126 readmitted encounters compared with 85,982 non-readmitted encounters.

The exploratory results identify previous inpatient utilization as the strongest numerical correlate of readmission and discharge disposition as the strongest categorical correlate. These findings provide an initial understanding of the data and inform the selection and evaluation of predictive models.

Limitations: The findings should be interpreted in light of the dataset's historical nature, the absence of some potentially important social and post-discharge information, and the substantial imbalance in the target variable. Multiple encounters may belong to the same patient, so patient-level splitting is preferable to reduce information leakage.

Furthermore, the current binary target combines readmissions within 30 days and readmissions after 30 days. If the intended outcome is specifically 30-day readmission, the target definition must be revised before the final predictive modeling stage.

Transition to Part 2: The cleaned and encoded datasets provide the basis for subsequent machine learning analysis. Part 2 will focus on model development and comparison, threshold selection, predictive performance, feature interpretability, error analysis, and subgroup fairness evaluation.

Fair, Interpretable, and Actionable Readmission Risk Modeling — Diabetes 130-US Hospitals: Machine Learning Analysis and Evaluation (Part 2)

1. Title and Brief Introduction

Title: Fair, Interpretable, and Actionable Readmission Risk Modeling — Diabetes 130-US Hospitals: Machine Learning Analysis and Evaluation (Part 2)

Introduction: This report details the second phase of a comprehensive project aimed at analyzing and predicting 30-day hospital readmissions among diabetic patients. Building upon the foundational data preparation, Part 2 focuses on the development, rigorous evaluation, and clinical interpretation of machine learning models. The analysis emphasizes that predictive modeling in healthcare extends beyond algorithmic accuracy, requiring careful consideration of threshold selection, error profiling, feature interpretability, and subgroup fairness to ensure safe and fair clinical decision support.

2. Objective of Part 2 and its Relationship to Part 1

Objective of Part 2: The primary goal of Part 2 is to construct, tune, and evaluate predictive models for 30-day readmission (target: readmitted == '<30') using the cleaned dataset. The workflow sets up interpretable baseline models, implements advanced ensemble techniques, and performs extensive post-hoc analyses—including threshold tuning, error analysis, and fairness evaluation—to figure out the clinical viability of the models.

Relationship to Part 1: Part 1 of the project managed the first data ingestion, exploratory data analysis, and preprocessing, culminating in the refined dataset (diabetic_cleaned.csv). Part 2 uses this cleaned dataset as its direct input, transitioning the project from data engineering to predictive modeling and clinical decision-support evaluation.

3. Methodology and Workflow

The analytical pipeline was executed using Python, leveraging libraries such as scikit-learn, XGBoost, and pandas. The method included the following key stages:

1. **Data Splitting:** The dataset was partitioned into training (80%), validation (10%), and test (10%) sets. Due to the absence of the `patient_nbr` identifier in the loaded dataframe, the pipeline fell back to an encounter-level split. It is noted that this approach may yield slightly optimistic metrics, as multiple encounters from the same patient could theoretically appear across different splits.
2. **Preprocessing:** Numerical features underwent median imputation and standard scaling. Categorical features were encoded using one-hot encoding with a minimum frequency threshold to manage dimensionality and prevent overfitting.
3. **Modeling Strategy:**
 - *Baselines:* Logistic Regression (with balanced class weights) and a shallow Decision Tree (max depth of 4) were set up to provide interpretable performance floors.
 - *Extended Models:* Random Forest and XGBoost were trained to capture complex, non-linear relationships without applying synthetic oversampling (SMOTE) or TF-IDF transformations.
4. **Threshold Tuning:** Recognizing the severe class imbalance (positive readmission rate of $\sim 11.5\%$), the default 0.50 probability threshold was abandoned. Instead, thresholds were tuned on the validation set to target a specific clinical recall (e.g., 0.60), framing the threshold as a clinical policy choice rather than a fixed algorithmic output.
5. **Evaluation and Post-Hoc Analysis:** Models were evaluated using Accuracy, Precision, Recall, F1-Score, ROC-AUC, PR-AUC, and Brier Score. Furthermore, error analysis was conducted to profile false negatives by length of stay and primary diagnosis. Subgroup fairness was assessed using out-of-fold predictions to ensure fair model performance across demographic groups.

4. Results and Analysis

Model Performance and Threshold Tuning: The extended models, particularly XGBoost, outperformed the baseline approaches. The tuned XGBoost model achieved a Test ROC-AUC of 0.671 (95% CI: 0.654–0.688) and a Test PR-AUC of 0.227 (95% CI: 0.205–0.248). The analysis concluded that using a default 0.50 threshold on imbalanced clinical data is clinically meaningless. By tuning the threshold to achieve a target recall of 0.60, the model's operating point was aligned with clinical priorities, explicitly balancing the trade-off between capturing true readmissions and managing false alarms.

Feature Importance and Clinical Interpretability: Both linear (Logistic Regression coefficients) and tree-based (XGBoost gain) feature importance analyses revealed consistent clinical drivers. The strongest correlates of 30-day readmission were prior inpatient use (number_inpatient), discharge destination (discharge_disposition_id), emergency-visit history, and overall use burden (num_medications, num_lab_procedures, time_in_hospital). Notably, diabetes-specific medication changes carried only a weak signal. This shows that the model primarily captures general hospital use and clinical complexity rather than isolated diabetes control metrics. Importantly, these top features are available at the time of discharge, requiring no post-discharge information or lab values.

Error Analysis: Aggregate recall metrics can obscure systematic blind spots. Error analysis of the test set revealed that false negatives cluster in specific subgroups, such as short-stay patients. This highlights a structural limitation: the current dataset lacks post-discharge signal, which is critical for predicting readmissions in patients with brief hospital stays. The error profiling confirms that the model serves best as a triage aid, showing where human review and targeted interventions are most needed.

Subgroup Fairness: Fairness evaluation showed that while the model performs adequately on average, recall varies across demographic subgroups (e.g., age groups, yielding a significant chi-square test result showing disparate recall rates). These variations suggest that a resource-allocation policy built directly on the model's raw outputs could deliver materially different follow-up rates to different populations, needing careful operational oversight.

5. Summary and Conclusions

Summary: Part 2 set up a robust, interpretable predictive framework for 30-day hospital readmissions. The analysis underscores that in clinical machine learning; the model's probability output is only half of the solution; the classification threshold is a critical policy choice that must be defined in collaboration with clinical stakeholders. The strongest predictors of readmission are rooted in historical healthcare use and discharge disposition, providing actionable insights for care teams at the point of discharge.

Limitations: The findings must be interpreted within the context of several limitations. The underlying data spans 1999–2008 and uses ICD-9 coding. It lacks social determinants of health, post-discharge information, and granular lab values (relying only on flags). Furthermore, the <30 readmission class stands for a small minority of encounters, inherently introducing noise into the estimates. No external validation has been conducted.

Final Recommendations: The developed models should be used strictly as triage aids to support clinical decision-making and **must not be used to deny care or to make autonomous**

discharge decisions: Before any operational deployment, the model's fairness metrics must be re-estimated on a contemporary cohort with adequate subgroup sample sizes. Any residual disparities must be resolved with clinical stakeholders, potentially through group-specific operating thresholds or by wrapping the model in a human-in-the-loop review protocol.